Stock Scanner — Product Design Doc

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Overview

Stock Scanner turns a pasted list of NSE stocks into one table of technical indicators, so a swing or positional trader can shortlist candidates in seconds instead of opening a chart per stock.

Today that check means flipping through TradingView charts one by one: RS vs Nifty, RSI on three timeframes, four moving averages, pivots and a judgement of trend stage. The scanner computes all of these the same way every time, side by side, sortable and exportable.

The first version is a personal web app that runs locally, uses free Yahoo Finance data and benchmarks every stock against the Nifty 50.

Goals and non-goals

The product succeeds if a 50-stock watchlist scans in under 30 seconds with values a trader trusts as matching their charting tool.

Goals

- Accept stocks as NSE symbols, ISINs or company names, mixed in one list.
- Return one row per stock with RS55, RSI (daily, weekly, monthly), SMA 9/20/50/200, classic pivot distance and Weinstein stage.
- Match TradingView's values for RSI and SMAs to within 0.5 points on liquid stocks.
- Make every derived label explainable: a stage or pivot shows the numbers behind it on hover.
- Let users sort, switch between distances and raw values, and export to CSV.

Non-goals for v1

- Charts, alerts, screening the whole market, or saved watchlists across devices.
- Intraday or real-time data; end-of-day and delayed prices are enough.
- BSE-only stocks, F&O data, fundamentals or news.
- Multi-user hosting, accounts or payments.

Users and use cases

The primary user is an individual Indian equity trader who follows momentum and stage-based methods (Weinstein, Minervini-style) and already reads these indicators on charts.

Use case	What the user does	What the table must make obvious
Evening watchlist review	Pastes 20–80 stocks after market close	Which are in Stage 2 with RS55 above 0 and rising
Entry timing	Sorts a shortlist by daily pivot distance or SMA 20 distance	Who is near support versus extended
Multi-timeframe check	Compares daily, weekly and monthly RSI	Whether short-term strength agrees with the longer trend
Weeding out weakness	Sorts by stage or RS55 ascending	Stage 4 names and Nifty underperformers to drop
Record keeping	Exports CSV	A dated snapshot to compare next week

Functional requirements

The app has one screen: an input box, a results table and a list of skipped inputs.

Input

- FR1. Accept up to 200 entries separated by commas, semicolons or new lines; spaces stay inside names.
- FR2. Resolve each entry in this order: explicit Yahoo ticker (`TCS.NS`, `500325.BO`), NSE symbol, ISIN, exact company name, name prefix, fuzzy name match.
- FR3. Drop duplicate entries, case-insensitively.
- FR4. Flag ambiguous or fuzzy matches on the row and list the other candidates.
- FR5. List every entry that could not be resolved or priced, with the reason, instead of failing the whole request.
- FR6. Remember the last input in the browser.

Table columns

Group	Column	Shown as
Stock	Symbol, company name	Text, with a warning icon for notes
Price	Close, day change	Value and %
Relative strength	RS55 vs Nifty 50	Signed ratio with an up/down arrow vs 5 sessions ago
RSI (14)	Daily, Weekly, Monthly	Value, colour-banded
SMA	9, 20, 50, 200	% from price, or the SMA value (toggle)
Classic pivot	Day, Week, Month	% from P, or the P level (toggle); R1/S1/R2/S2 on hover
Stage	Weinstein stage	Badge with qualifier; reasoning on hover

Interactions

- FR7. Click any column header to sort; click again to reverse. Missing values sort last.
- FR8. Toggle SMA and pivot columns between distance and level without refetching.
- FR9. Export all rows, with both values and distances, to CSV.
- FR10. Show the Nifty 50 close and the data date the results are based on.

Indicator definitions

Every indicator is computed from split- and bonus-adjusted daily OHLCV bars; weekly and monthly bars are built from them, never fetched separately.

Bars. Weeks end on Friday and months on the last calendar day. Open is the first open, high the max, low the min, close the last close, volume the sum. The current week and month are kept as live, incomplete bars, as TradingView does.

RS55. Stock return over 55 sessions divided by Nifty 50 return over the same sessions, minus one. Only dates present in both series are used. Above 0 means the stock beat the index; the arrow compares today with 5 sessions ago.

```
RS55_t = \frac{C_t / C_{t-55}}{I_t / I_{t-55}} - 1
```

RSI (14), Wilder. Applied separately to daily, weekly and monthly closes. The first average gain and loss are simple means of the first 14 changes; each later value is smoothed. If the average loss is 0, RSI is 100 (or 50 when both averages are 0).

```
\bar{G}_t = \frac{13\,\bar{G}_{t-1} + G_t}{14}, \quad \bar{L}_t = \frac{13\,\bar{L}_{t-1} + L_t}{14}, \quad RSI_t = 100 - \frac{100}{1 + \bar{G}_t / \bar{L}_t}
```

Five years of history is loaded so the recursive smoothing has settled; monthly RSI on fewer than 40 months carries a warning.

SMA 9, 20, 50, 200. The plain mean of the last N daily closes. Distance from price = (Close ÷ SMA − 1) × 100.

Classic pivot. Levels in force for the current day, week and month, from the previous completed bar of that timeframe. Distance = (Close ÷ P − 1) × 100.

```
P = \frac{H + L + C}{3}, \quad R1 = 2P - L, \quad S1 = 2P - H, \quad R2 = P + (H - L), \quad S2 = P - (H - L)
```

Weinstein stage. Based on the 30-week SMA of weekly closes and its slope, the % change over the last 4 weeks. A slope within ±1% counts as flat.

MA slope	Price vs 30-week MA	Stage
Rising	Above	2 Advancing; "early" if the MA was not rising 8 weeks ago
Rising	Up to 5% below	2 (pullback)
Rising	More than 5% below	3 Topping
Falling	Below	4 Declining; "early" if the MA was not falling 8 weeks ago
Falling	Up to 5% above	4 (bounce)
Falling	More than 5% above	1 Basing
Flat	Either	3 if the MA rose over the prior ~6 months, else 1

A Stage 2 stock with RS55 below 0 gets a caution in its reasoning. All thresholds (30 weeks, 4-week slope, ±1%, 5%) are configurable.

UX design

The screen is a dense, scannable table: colour marks state, and detail lives in hover text rather than extra columns.

Layout. Title and one-line description, then the input box with Analyze and Export CSV beside it. A toolbar holds the SMA and pivot toggles and the data date. The table follows, with grouped headers (RSI, SMA, Classic pivot) and a sticky header row. A short legend under the table restates each formula in one line.

Colour coding

Element	Rule
RSI	70 and above hot (amber), 60–70 strong (green), 40–60 neutral, 30–40 weak (red), 30 and below cold (blue)
RS55	Green chip above 0, red below
SMA and pivot distance	Green when price is above, red when below
Stage badge	1 neutral sand, 2 green, 3 orange, 4 red

Colour is never the only signal: values keep their sign and stages keep their number and name. Light and dark themes follow the system setting.

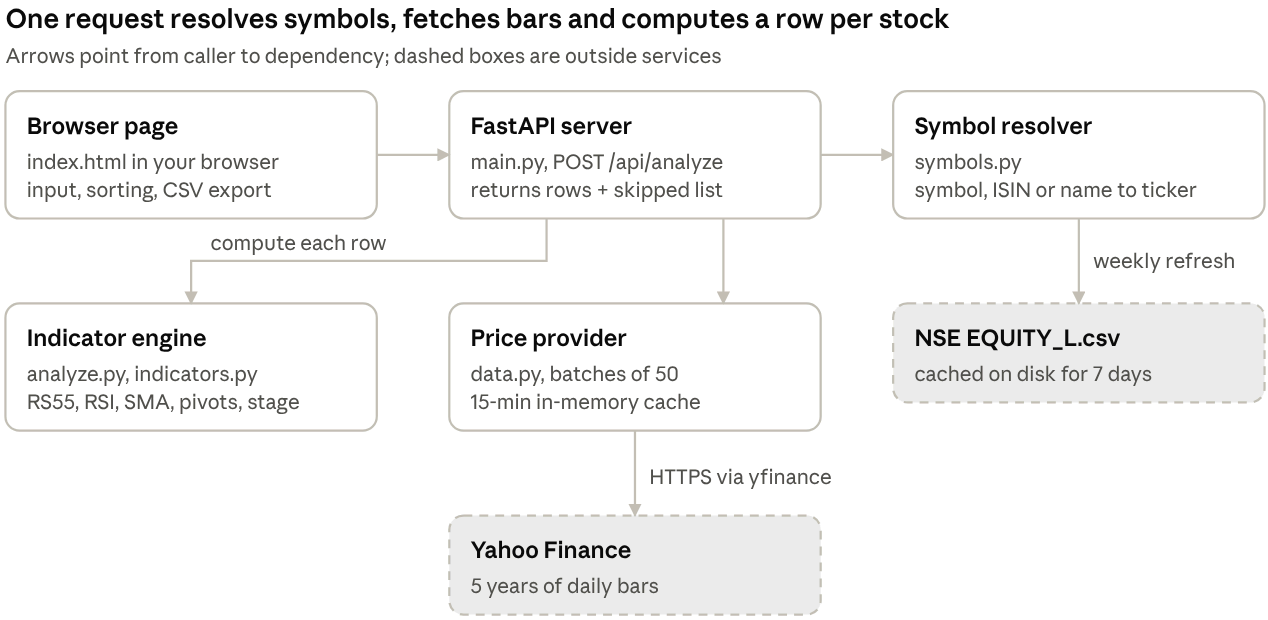
Hover text. SMA cells show the SMA value; pivot cells show P, R1, S1, R2, S2 and the source bar's date; stage badges show the reasoning sentence; the warning icon shows match notes and data warnings.

States

- Empty: a prompt to enter stocks.
- Loading: the Analyze button reads "Loading…" and is disabled.
- Partial success: rows render, and a red strip lists skipped entries with reasons.
- Failure (benchmark unavailable, bad input): an error line under the input; the previous table stays.
- Short history: missing values show as a dash, with the reason in the row's warning icon.

System design

A local Python server does all the work; the browser page only sends the list and renders the rows it gets back.



architecture · 6 components, 2 outside services

The indicator engine has no network access, so every formula is unit-tested on fixed data, and the price provider can be replaced without touching it.

Stack. Python 3.10+, FastAPI and Uvicorn, pandas and NumPy, yfinance; a single HTML page with plain JavaScript, no build step.

Data source. Yahoo Finance via yfinance, five years of daily bars per ticker, Nifty 50 as `^NSEI`. Prices use Yahoo's `Close`, which is split- and bonus-adjusted but not dividend-adjusted, matching TradingView's default. The provider is an interface with one method, `history(tickers)`, so a broker API can replace it.

Caching. Price frames are cached in memory for 15 minutes per ticker, so only tickers not already cached are downloaded. The NSE equity list is cached on disk and refreshed every 7 days, falling back to the stale copy, and then to symbol-only mode if it was never downloaded.

API contract

`POST /api/analyze` takes `{"symbols": "RELIANCE, TCS\nTata Motors"}` (a string or a list) and returns:

Field	Contents
`rows[]`	One object per stock: `symbol`, `name`, `close`, `change_pct`, `rs55`, `rs55_5d_ago`, `rsi_d/w/m`, `sma{9,20,50,200}` and `_dist`, `pivot_{d,w,m}` (P, R1–R3, S1–S3, `from`) and `_dist`, `stage` (number, name, label, reason), `warnings[]`, `note`
`problems[]`	Skipped entries: `query` and `reason`
`benchmark`	Nifty 50 ticker, close and last date
`generated_at`	UTC timestamp

Errors: 400 for empty input or more than 200 entries; 502 when Nifty 50 data cannot be downloaded. NaN values are sent as `null`.

Non-functional requirements

Area	Requirement
Speed	50 stocks in under 30 s on first load; under 1 s when cached
Accuracy	RSI and SMA within 0.5 of TradingView on liquid NSE stocks; RSI verified against a Wilder reference in tests
Resilience	One bad ticker or calculation never fails the request; it is listed as skipped
Limits	200 entries per request; downloads batched 50 tickers at a time
Freshness	Price cache expires after 15 minutes; NSE symbol list refreshed weekly, stale copy used if the refresh fails
Testability	Indicator code is pure (no network) and covered by unit tests; API tested with a fake price source
Portability	Python 3.10+, runs on Windows, macOS and Linux with one install command

Area	Requirement
Privacy	Runs locally; inputs are stored only in the user's browser

Risks, open questions and roadmap

The biggest risk is the data source: yfinance is unofficial and can break without notice.

Risk	Impact	Mitigation
Yahoo changes its endpoints	Scans fail until yfinance is patched	Pin and update yfinance; swap in a broker API behind the provider interface
Yahoo rate limits the IP	Large or frequent scans get blocked	15-minute cache, batched downloads, 200-entry cap
Unadjusted corporate actions	A bonus or split distorts SMAs and RSI for a few days	Show the data date; spot-check against TradingView; broker data later
Stage rules disagree with a chartist	Users lose trust in the label	Show the reasoning on hover; make thresholds configurable
Yahoo terms of use	Sharing or selling the app is not allowed on this data	Keep v1 personal; license data before any public release

Open questions

- [ ] Should the stage thresholds (±1% slope, 5% band) be tuned against a set of hand-labelled charts?
- [ ] Is RS55 alone enough, or should Mansfield RS (52-week) be added?
- [ ] Should weekly and monthly pivots switch to the next period after Friday's close and month-end?

Roadmap

1. v1 (built): local app, yfinance, the full indicator set, sorting and CSV export.
2. v1.1: saved watchlists, configurable thresholds in the UI, a mini price chart on row hover.

3. v2: broker API data (Angel One SmartAPI or Kite Connect), scheduled end-of-day scans, alerts on stage changes.
4. v3: hosted multi-user version on licensed data.